Semi-Supervised Image Classification with MixMatch and FixMatch on CIFAR-10

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Abstract

We implement and test two semi-supervised learning (SSL) methods, MixMatch [1] and FixMatch [2], on CIFAR-10 using a small CNN. We look at how accuracy changes with the number of labeled examples ($N_\ell \in \{250, 500, 1000, 5000\}$) and test how sensitive FixMatch is to the confidence threshold $\tau \in \{0.70, 0.90, 0.95\}$ as well as how different augmentation strategies perform. FixMatch with τ=0.70 gets **58.15%** test accuracy at N_ℓ=1000, which beats both the supervised baseline (54.56%) and MixMatch (42.93%). We show that MixMatch performs poorly because the λ_U ramp-up never activates in our short training run, and that FixMatch works better with a lower threshold when using a weaker model.

1. Introduction

Training deep learning models usually requires a lot of labeled data. In practice, getting labels is expensive while unlabeled data is easy to collect. Semi-supervised learning (SSL) tries to solve this by using both labeled and unlabeled data during training.

MixMatch [1] and FixMatch [2] are two well-known SSL methods for image classification. Both are tested on CIFAR-10 and use data augmentation together with consistency regularisation to get information out of unlabeled images. In this project we implement both and look at two things:

1. How does accuracy change with the number of labeled examples?
2. How sensitive is FixMatch to the confidence threshold τ?

We compare results against a supervised baseline trained on the same labeled data and against a fully supervised upper bound (about 73.5% with 45 000 labels).

2. Background

2.1. MixMatch

MixMatch [1] combines several SSL ideas into one training loop. Given a labeled batch $\mathcal{X}$ and unlabeled batch $\mathcal{U}$:

1. Each unlabeled image is augmented K times and the model's softmax outputs are averaged to get a label guess: $\bar{q}_b = \frac{1}{K} \sum_k p_m(y \mid \alpha_k(u_b))$.
2. The guess is sharpened using temperature T: $\hat{q}_{b,c} \propto \bar{q}_{b,c}^{1/T}$.
3. All labeled and unlabeled samples are shuffled together into $\mathcal{W}$ and mixed with MixUp [3], using $\lambda' = \max(\lambda, 1-\lambda)$ to keep each sample closer to its original.

4. The total loss is $\mathcal{L} = \mathcal{L}_X + \lambda_U(t)\,\mathcal{L}_U$ where $\mathcal{L}_X$ is soft cross-entropy on labeled data and $\mathcal{L}_U$ is MSE on unlabeled data. $\lambda_U(t)$ is linearly increased over 16 000 steps.

Paper hyperparameters for CIFAR-10: T=0.5, K=2, α=0.75, λ_U=75, Adam with η=0.002.

2.2. FixMatch

FixMatch [2] uses pseudo-labels with a confidence filter. For each unlabeled image u_b:

1. A *weak* augmentation gives a soft prediction $q_b = p_m(y \mid \alpha(u_b))$.
2. If $\max(q_b) \geq \tau$, the argmax becomes a hard pseudo-label $\hat{q}_b = \arg\max(q_b)$.
3. A *strong* augmentation of the same image is trained against $\hat{q}_b$:

$$\mathcal{L}_U = \frac{1}{\mu B} \sum_b \mathbf{1}[\max(q_b) \geq \tau]\; H(\hat{q}_b,\; p_m(y \mid \mathcal{A}(u_b)))\,.$$

Total loss: $\mathcal{L}_X + \lambda_u\,\mathcal{L}_U$ with λ_u=1. Paper defaults: τ=0.95, μ=7, SGD with Nesterov momentum and cosine LR decay.

3. Experimental Setup

3.1. Dataset

CIFAR-10 [4] has 60 000 32×32 RGB images in 10 classes (50 000 train / 10 000 test). We hold out 5 000 training images as a validation set using stratified sampling. The labeled subset is picked with balanced class counts from the remaining 45 000 images.

3.2. Model Architecture

We use a small CNN with two convolutional blocks and two fully connected layers (about 580 K parameters):

$$\underbrace{\text{Conv}(3{\to}32)}_{\text{ReLU, MaxPool}} \;\to\; \underbrace{\text{Conv}(32{\to}64)}_{\text{ReLU, MaxPool}} \;\to\; \underbrace{\text{FC}(4096{\to}128)}_{\text{ReLU, Dropout}} \;\to\; \text{FC}(128{\to}10)$$

This model is fast enough to train on CPU and still performs reasonably on CIFAR-10.

3.3. Augmentations

Weak: random horizontal flip and random crop with padding 4. Used for all labeled data and unlabeled weak views.
Strong (FixMatch variations): All strong augmentations begin with the weak baseline steps, followed by one of five specialized distortion strategies:

- RandAugment: Applies two randomly sampled operations at magnitude M=9 from a standard pool of geometric and color transformations.
- AutoAugment: Utilizes a fixed, preset policy optimized specifically for the CIFAR-10 dataset distribution.
- Cutout: Concatenates a random rectangular erasing mask onto a single area of the image to occlude local spatial features. This is added as another augmentation layer after using RandAugment as proposed by [2].
- TrivialAugmentWide: Uniformly samples a single transformation from a wide, continuous range of magnitudes without hyperparameter tuning.
- AugMix: Blends multiple parallel transformation chains together with the original image using a Dirichlet distribution to preserve structural consistency.

3.4. Training Details

- **Optimizer:** Adam, η=0.001, weight decay 5×10^{-4}, 20 epochs.
- **MixMatch:** B_ℓ=B_u=64. Steps per epoch defaults to the length of the labeled loader (about 4 to 78 steps depending on N_ℓ). The λ_U ramp-up length is fixed at 16 000 steps as in the paper.
- **FixMatch:** B_ℓ=64, B_u=128, `steps_per_epoch` $\in\{100, 400\}$, τ varied.
- **Supervised baseline:** same model, 400 steps/epoch, trained only on labeled data.

All experiments use seed 0.

4. Results

4.1. MixMatch: Effect of Labeled Data Amount

Figure 1 shows validation accuracy over 20 epochs for $N_\ell \in \{250, 500, 1000, 5000\}$. All settings improve over time. With 5000 labels the model reaches 54.24% test accuracy and with 250 labels it reaches 28.92%.

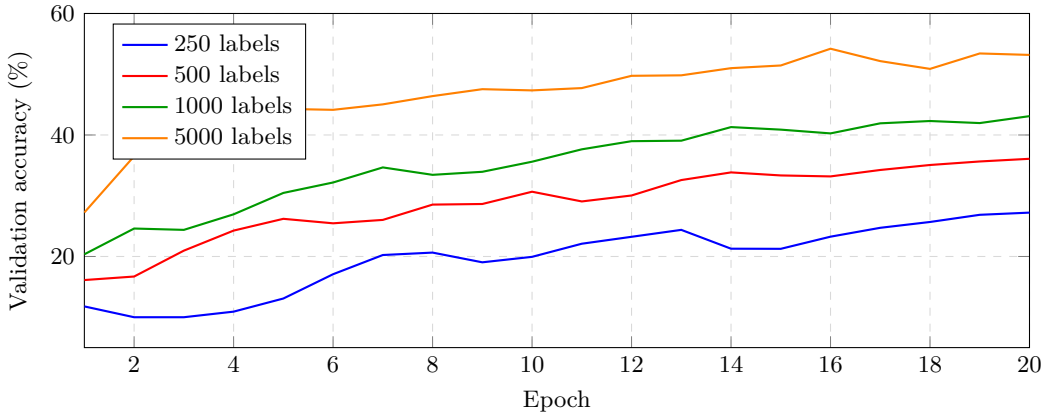


Figure 1: MixMatch validation accuracy over 20 epochs for different labeled set sizes. More labels consistently gives faster and higher accuracy.

4.2. Method Comparison at 1000 Labels

Table 1 and Figure 2 compare all methods at N_ℓ=1000.

Table 1: Test accuracy at N_ℓ=1000 labeled examples on CIFAR-10.

Method	Config	Steps/epoch	Best val acc	Test acc
Supervised baseline	N/A	400	55.44%	54.56%
MixMatch	N/A	∼15	43.10%	42.93%
FixMatch	$\tau = 0.70$	400	58.64%	**58.15%**
FixMatch	$\tau = 0.70$	100	55.14%	55.86%
FixMatch	$\tau = 0.90$	100	53.86%	54.48%
FixMatch	$\tau = 0.95$	100	54.18%	54.73%
Supervised (45 000)	N/A	N/A	N/A	≈73.48%

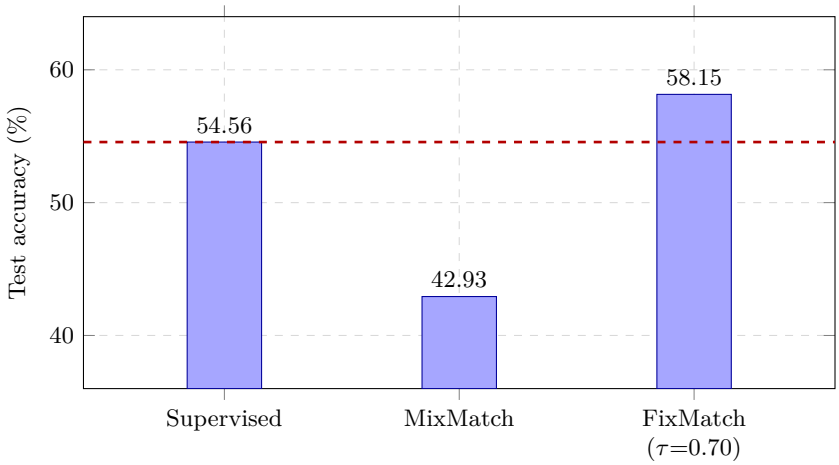


Figure 2: Test accuracy at N_ℓ=1000. The dashed line is the supervised baseline. FixMatch (τ=0.70, 400 steps/epoch) is the only method that beats it.

4.3. FixMatch: Threshold Sensitivity

Figure 3 shows how τ affects accuracy (all runs: 100 steps/epoch, 20 epochs, N_ℓ=1000). A lower τ lets more unlabeled samples contribute pseudo-labels, which helps when the model is weak.

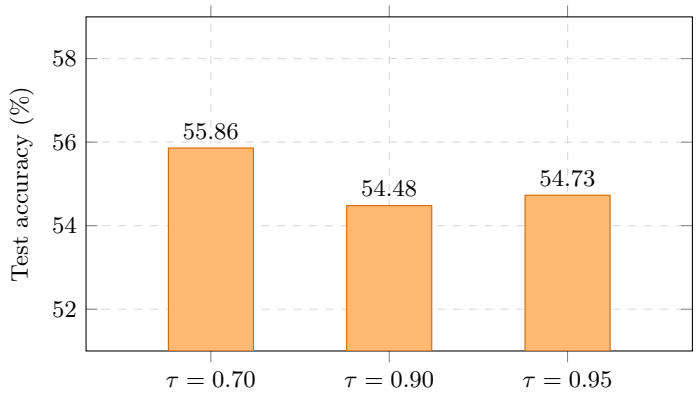


Figure 3: FixMatch test accuracy for different confidence thresholds τ (N_ℓ=1000, 100 steps/epoch, 20 epochs). Lower τ works best for the small CNN.

4.4. FixMatch Training Dynamics

Figure 4 shows validation accuracy and the fraction of unlabeled samples used (τ=0.80, 400 steps/epoch) over 20 epochs. In epoch 1 only 2.2% of unlabeled samples pass the threshold. By epoch 20 this grows to about 46%. This shows how FixMatch builds on itself: as the model gets better, more unlabeled samples pass the threshold and contribute to training.

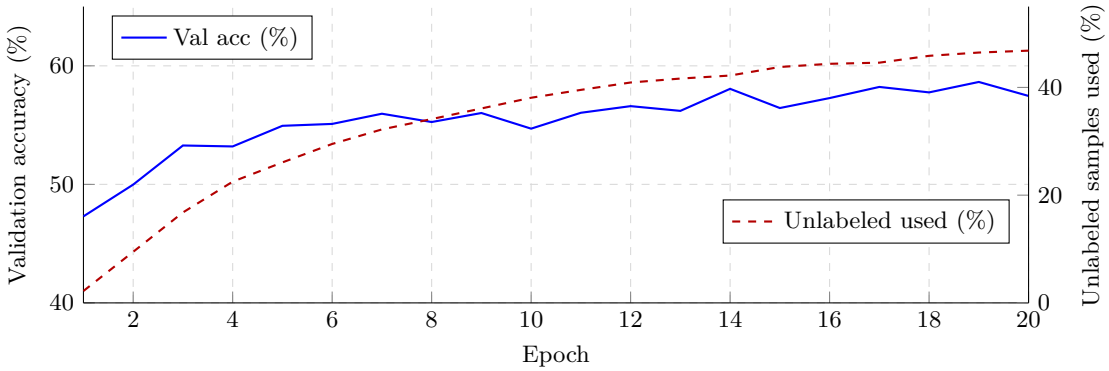


Figure 4: FixMatch training dynamics (τ=0.80, 400 steps/epoch, N_ℓ=1000). More unlabeled samples pass the confidence threshold as training progresses (right axis).

4.5. FixMatch: Augmentation Strategy Ablation

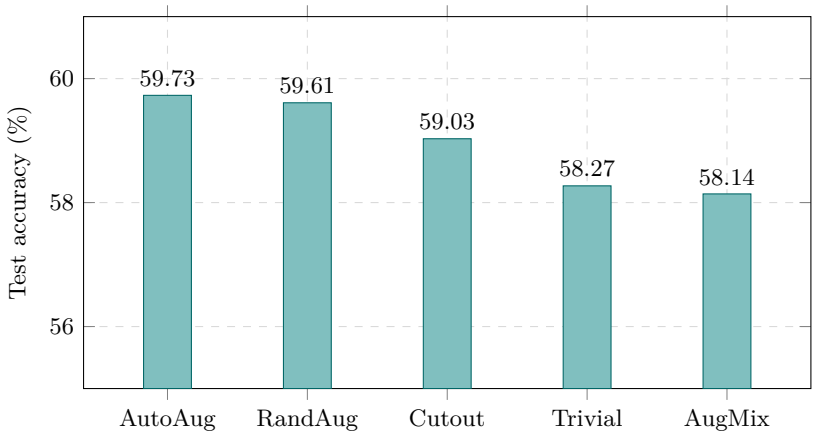


Figure 5

Figure 5 evaluates how five distinct strong augmentation strategies affect final test accuracy (N_ℓ=1000, τ=0.80, 400 steps/epoch). All variations demonstrate strong performance, clustering within a narrow window.

FixMatch Augmentation Strategy Analysis (Labels: 1000 | Steps: 400 | Threshold: 0.8)

Augmentation Strategy	Test Acc	Time (s)	Final Unlabeled Usage
AutoAugment	0.5973	927.0	0.4728
RandAugment	0.5961	947.3	0.4631
Cutout	0.5903	996.8	0.4688
Trivial	0.5827	881.7	0.4819
AugMix	0.5814	2055.1	0.4689

Figure 6: Raw results from implementing different augmentation startegies with FixMatch.

5. Discussion

MixMatch underperforms the supervised baseline. With N_ℓ=1000 and 20 epochs, MixMatch (42.93%) is worse than the supervised baseline (54.56%). The reason is the λ_U ramp-up: the paper ramps λ_U from 0 to 75 over 16 000 steps, but because steps per epoch defaults to the labeled loader length (about 16 steps at N_ℓ=1000), the full run only covers around 320 steps. This means λ_U only reaches about $75 \times 320/16000 \approx 1.5$ at the end of training. The unlabeled loss never really contributes (Table 2), so MixMatch effectively trains as a supervised model with extra noise. Running more steps or shortening the ramp-up would fix this.

Table 2: MixMatch loss components at selected epochs (N_ℓ=1000). $\mathcal{L}_U$ stays very small because λ_U never ramps up properly.

Epoch	$\mathcal{L}$	$\mathcal{L}_X$	$\mathcal{L}_U$
1	2.297	2.297	0.0026
10	2.042	2.036	0.0083
20	1.917	1.901	0.0113

FixMatch works better with a lower threshold. The paper uses τ=0.95 but we find τ=0.70 works best with our small CNN. A weaker model gives lower confidence scores overall, so a high threshold cuts out nearly all unlabeled data early in training. As seen in Figure 4, even with τ=0.80 only 2.2% of unlabeled data is used in epoch 1. With τ=0.95 this would be even lower. Choosing τ based on model size matters.

FixMatch beats the supervised baseline. FixMatch with τ=0.70 and 400 steps/epoch gets 58.15%, which is 3.6 percentage points above the supervised baseline (54.56%) using the same 1000 labeled examples. This shows that FixMatch can make good use of unlabeled data even with a small model.

Why augmentation diversity outpaces blending and masking. AutoAugment and RandAugment may be performing the best because their diverse distortions prevent overfitting without destroying the core structural context of the image. Conversely, Cutout could be reducing accuracy because its hard masks risk completely erasing critical diagnostic features in low-resolution (32×32) images.

While AugMix maintains high unlabeled data usage, its parallel blending chain acts as a low-pass filter that blurs sharp edge details. Learning from these complex, blurry mixtures requires massive computational overhead and proves too difficult for a small CNN's limited capacity.

Gap to full supervision. All SSL methods are far from the fully supervised upper bound (about 73.5% with 45 000 labels). This is expected since we use a short training budget and a small model. The original papers use WRN-28-2 and much longer training, where SSL gets close to fully supervised performance with as few as 250 labels.

6. Conclusion

We implemented MixMatch and FixMatch on CIFAR-10 and found that FixMatch beats the supervised baseline when using τ=0.70, reaching 58.15% with 1000 labels. MixMatch did not benefit from unlabeled data because the λ_U ramp-up never activated in our short training run. More labeled data helps MixMatch consistently across 250 to 5000 labels. Future work should train both methods longer and shorten the ramp-up schedule to match the actual number of steps, and test the extra augmentation strategies (colour jitter, Gaussian blur and rotation) that we also implemented.

References

[1] D. Berthelot, N. Carlini, I. Goodfellow, N. Papernot, A. Oliver and C. Raffel. MixMatch: A holistic approach to semi-supervised learning. In *NeurIPS*, 2019.

[2] K. Sohn, D. Berthelot, C.-L. Li, Z. Zhang, N. Carlini, E. Cubuk, A. Kurakin, H. Zhang and C. Raffel. FixMatch: Simplifying semi-supervised learning with consistency and confidence. In *NeurIPS*, 2020.

[3] H. Zhang, M. Cisse, Y. Dauphin and D. Lopez-Paz. mixup: Beyond empirical risk minimization. In *ICLR*, 2018.

[4] A. Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.